Income Imputation for the Canadian Community Health Survey

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1. Introduction

Income questions are asked during the Canadian Community Health Survey (CCHS) interview because of their potential association with health variables. The income variable most commonly used in analyses is total household income (variable INC_3). However, because of the sensitive nature of reporting income, only 66% of the CCHS respondents reported their income in 2011. For nonrespondents to total household income, a series of questions (INC_5A/B/C) are asked to determine a range for the income. In 2011, 45% of income nonrespondents provided the range. Please refer to the CCHS questionnaire for more information on the income module.

In addition to the low response rate, a study conducted in 2009 suggested that the income nonrespondents had different health characteristics than the respondents. Because of this difference in health profiles and the low response rate, analyses that are based exclusively on income respondents may be biased. It was therefore decided to impute total household income starting with the 2011 CCHS release.

For users interested in conducting analyses focusing specifically on household income, it is highly recommended to rely on other income data sources rather than using the CCHS. Those interested in analysing household income as it relates to health should use derived income variables such as income quintiles rather than the actual income value directly. The imputation process performs well at predicting income quintiles but there is variability in predicting the exact income value.

2. Modeled Household Income

An important auxiliary variable used in the imputation process is **modeled household income.** It is computed for all CCHS respondents (whether they responded to the income questions or not) using a three step process. Details are given below.

<u>Step 1</u>
Using a regression model, personal income of *all* members in a responding household is derived. Variables used in the model include age group, education level, sources of income, province of residence, marital status, household size, whether the dwelling is owned or rented, and whether the person selected for CCHS interview is an adult. It is worthwhile to note that no health variables are used in the model in order to prevent creating artificial relationship between income and certain health variables.

Step 2
A preliminary household income value is derived by summing all personal incomes from step 1. The resulting totals do not accurately reflect the income information collected by the CCHS so these values require further adjustments.

Step 3
The median household income from T1 tax data is derived for each postal code by household size group. Using a regression model with the median income from tax and the preliminary household income from step 2, a final modeled household income is derived. This final modeled household income does not appear on the final data file, but is rather used as an auxiliary variable in the nearest neighbour imputation process.

3. Imputation Process

A nearest neighbour imputation method is used to impute income. The modeled household income defined above is used as a distance measure to determine which pair of donor-recipient is the "nearest" within imputation classes.

Imputation classes are formed differently depending on whether the income nonrespondents provide an income range or not. For those who provide an income range, imputation classes are based on the income range and household size. Since the income ranges are small, incomes imputed using this approach are very accurate. For people who do not provide an income range, imputation classes are formed based on health region and household size. Even though these imputed income values are not as accurate as those imputed using the income range, their quality is still acceptable. The following table shows the results of imputation for the 2011 CCHS Master file:

Imputation Step	**No. of people**	**Percent of the total**
Reported income	42,215	66.4%
Imputation within income range by household size	9,613	15.1%
Imputation within health region by household size	11,714	18.4%
Total	**63,542**	**100%**

4. 2011 Data File

Beginning with the release of 2011 data, almost all respondents have a value for total household income (INC_3). The only exception is for 'partial share' units on the share file. In some cases, a person most knowledgeable (PMKPROXY) was selected to provide the income information. When this person refuses to have this information included on the share file but the respondent agrees to share their information, a partial share

exists. In this case, the variable INC_3 is considered as 'not stated'. An imputation flag INCFIMP is included on the file to identify which records have an imputed value for total household income. In addition, derived income variables such as INCDHH, INCDADR, INCDRCA, etc., will be based on the imputed household income value.

It should be noted that total personal income (variable INC_8A) is not imputed so users should take the appropriate steps while analyzing this variable.